

3. API Contract & Error Handling

API Endpoint

GET /api/dashboard

Purpose:

Retrieves unified analytics data from Shopify and TikTok Shop for the merchant dashboard.

Example Request

JSON

```
{
  "merchantId": "MERCHANT_123",
  "dateRange": "7d"
}
```

Example Success Response

JSON

```
{
  "summary": {
    "totalSales": 12500,
    "orders": 245,
    "inventoryRisk": 12,
    "revenue": 8300
  },
  "platformStatus": {
    "shopify": {
      "status": "connected",
      "lastUpdated": "2026-06-23T10:00:00Z"
    },
    "tiktok": {
      "status": "delayed",
      "lastUpdated": "2026-06-23T09:45:00Z"
    }
  }
}
```

```
    }  
  },  
  
  "salesChart": [  
    { "date": "2026-06-17", "sales": 1200 },  
    { "date": "2026-06-18", "sales": 1800 },  
    { "date": "2026-06-19", "sales": 1500 }  
  ],  
  
  "inventory": [  
    {  
      "product": "Hoodie A",  
      "shopifyStock": 50,  
      "tiktokStock": 45,  
      "status": "synced"  
    }  
  ],  
  
  "orders": [  
    {  
      "id": "#1001",  
      "platform": "Shopify",  
      "customer": "John",  
      "amount": 50  
    }  
  ]  
}
```

TikTok API Failure Handling

Scenario:

TikTok API request fails while Shopify API request succeeds.

Example Response

JSON

```
{
  "summary": {
    "totalSales": 9800,
    "orders": 180,
    "inventoryRisk": 8,
    "revenue": 6200
  },

  "platformStatus": {
    "shopify": {
      "status": "connected",
      "lastUpdated": "2026-06-23T10:00:00Z"
    },

    "tiktok": {
      "status": "error",
      "message": "Unable to fetch TikTok data",
      "lastSuccessfulSync": "2026-06-23T08:30:00Z"
    }
  },

  "inventory": [
    {
      "product": "Hoodie A",
      "shopifyStock": 50,
      "tiktokStock": null,
      "status": "unavailable"
    }
  ],

  "orders": [
```

```
{
  "id": "#1001",
  "platform": "Shopify",
  "customer": "John",
  "amount": 50
}
]
```

Frontend Behavior

1. Dashboard continues to load normally.
2. Shopify data remains visible.
3. TikTok sections display a warning badge.
4. Inventory rows with missing TikTok data display "N/A".
5. A warning banner informs users that TikTok data is temporarily unavailable.

Design Rationale

The backend aggregates Shopify and TikTok data independently. Failure of one platform should not prevent the dashboard from rendering available data from the other platform. This ensures fault tolerance and a better user experience.